

1. Administrative & Study Identification

Full Study Title: A Phase I Randomized, Single-blind, Placebo-controlled Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of AZD6234 Following Repeat Dose Administration in Participants with Overweight or Obesity **Short Title/Acronym:** Repeat Dose Study of AZD6234 in Overweight/Obese Participants **Protocol Number:** D8750C00002 / IRAS ID: 1009332 **NCT Number:** NCT06132841 **Sponsor Name:** AstraZeneca AB **Investigational Product:** AZD6234 (Long-acting amylin analogue) **Phase of Development:** Phase 1 **Medical Monitor:** Johan Holmgren, AstraZeneca

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of AZD6234 after repeated subcutaneous (SC) dose administration compared to placebo in healthy male and female participants who are overweight or obese. **Secondary Objectives:** To evaluate the pharmacokinetics (PK), pharmacodynamics (PD) (including effects on body weight and insulin levels), immunogenicity, and the absorption, metabolism, distribution, and excretion profile of AZD6234.

2.2 Background & Rationale Obesity is defined as having a BMI of ≥ 30 kg/m² and is a major risk factor for the development of many other conditions, including type 2 diabetes mellitus, heart disease, and chronic kidney disease. This Phase 1 trial aims to test the clinical safety, tolerability, and early efficacy signals of AZD6234 as a potential new pharmacological treatment to aid in weight loss and the prevention of obesity-related comorbidities.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Single-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target N: 142 participants **Number of Sites:** Multicenter **Estimated Study Start:** Q4 2023 **Estimated Primary Completion:** H1 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy male and female participants aged 18 to 55 years inclusive with suitable veins for cannulation or repeated venipuncture. 2. Have a BMI between 25 and 40 kg/m² inclusive (at the time of screening) and weigh at least 60 kg. 3. Females must have a negative pregnancy test, not be lactating, and must be of non-childbearing potential (e.g., postmenopausal for ≥ 12 months, or irreversible surgical sterilization) or use adequate protection for specific cohorts.

4.2 Exclusion Criteria 1. Previous or planned (within study period) bariatric surgery or fitting of a weight loss device (e.g., gastric balloon or duodenal barrier). 2. Participants who follow a vegan diet or have medical dietary restrictions. 3. Vulnerable participants (e.g., kept in detention, protected adults under guardianship) or those who cannot communicate reliably with the Investigator.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Repeated doses of AZD6234 or matching placebo. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Screening period of up to 35 days, followed by the dosing phase and safety follow-up extending to Day 78.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications not affecting weight or metabolism, approved by the investigator. **Prohibited:** Exogenous hormonal treatments (for postmenopausal validation), oral contraceptives (in specific cohorts like Cohort 4 where other protection is required), and concurrent weight loss medications or devices.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -35 to -3	Informed Consent, Medical History, BMI/Weight verification, Pregnancy Labs
Baseline	Day 1	Randomization, First SC Dose, Clinical Unit Admission
Interim	Dosing Period	Subcutaneous injections, Safety Labs, PK/PD blood sampling (insulin levels, immunogenicity)
End of Study	Up to Day 78	Final Vital Signs, Body Weight Assessment, Exit Interview, Final Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability of repeat doses of AZD6234 (measured by the incidence of adverse

events, vital sign changes, and clinical laboratory abnormalities). **Measurement Method:** Clinical observation, patient reporting, and centralized laboratory safety assessments.

7.2 Secondary & Exploratory Endpoints Pharmacokinetics (PK) of AZD6234 (absorption, distribution, metabolism, excretion). Pharmacodynamics (PD), including percentage change in body weight and modulation of insulin levels. Immunogenicity (presence of anti-drug antibodies).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via direct observation in the Clinical Unit and patient reporting during follow-up visits. **Serious Adverse Events (SAEs):** Standard Phase 1 reporting timeline, typically within 24 hours to the Sponsor and Health Research Authority. **Data Monitoring Committee (DMC):** Internal safety review committees evaluate ascending dose cohort data prior to advancing exposure.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of the investigational product). **Sample Size Justification:** $N=142$ is suitably powered for a Phase 1 multiple ascending dose trial to identify safety signals, PK/PD parameters, and tolerability. **Handling of Missing Data:** Standard Phase 1 imputation rules for missing pharmacokinetic and pharmacodynamic data points.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict onsite monitoring in dedicated Phase 1 clinical research units. **Audit Trail:** E-CRF data entry, core laboratory review for any imaging (e.g., MRI for Cohort 4), and rigorous source data verification.

11. Study Identifiers & Governance

Primary ID: D8750C00002 **Registry Number:** NCT06132841 **Sponsor/Lead Organization:** AstraZeneca AB **Study Title:** A Study to Assess the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of AZD6234 After Repeat Dose Administration in Participants Who Are Overweight or Obese **Official Regulatory Title:** A Phase I Randomized, Single-blind, Placebo-controlled Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics

12. Clinical Framework (Obesity Specific)*

Primary Condition: Obesity, Diabetes Prevention, Hypertriglyceridemia **Diagnosis Threshold:** BMI between 25 and 40 kg/m2, weight ≥ 60 kg **Medical Methodology:** Pharmacological weight management intervention **Optimization Target:** Significant body weight reduction and insulin regulation (Note: Adapted from template's Hyperkalemia-specific placeholders to align with the actual NCT06132841 clinical context)

13. Study Procedures & Methodology

Management Pathway: Phase 1 clinical trial unit standardized pathway **Education Protocol (HCP):** Phase 1 protocol adherence and safety signal detection training **Education Protocol (Patient):** Clinical unit confinement rules, contraception compliance, and dietary restriction tracking **Quality Audit Mechanism:** Core laboratory review (e.g., MRI evaluations) and rigorous investigator oversight

14. Observation & Follow-Up Schedule

Total Duration: Up to 113 days (Max 35 days screening + 78 days treatment/follow-up) **Onsite Visit Cadence:** Frequent Phase 1 inpatient/outpatient visits through Day 78 **Remote Monitoring Frequency:** N/A (Primarily handled via intensive onsite tracking) **Checklist Review Interval:** Continuous during clinical unit admissions and scheduled follow-up checkpoints

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	Johan Holmgren	____	02-Apr-2026				
Clinical Operations Lead	[AstraZeneca ClinOps]	____	02-Apr-2026				
Lead Statistician	[AstraZeneca Biometrics]	____	02-Apr-2026				